

## 习 题 8

1.  $\alpha = (a_1, a_2), \beta = (b_1, b_2)$

(1) 没有正定性.

(2) 是, 根据定义逐一验证.

(3) 不是,  $\alpha = 0$  时,  $(\alpha, \alpha) = 1 \neq 0$ .

2. (1) 不是, 取  $\alpha = \beta = (1, 0, 0, \dots, 0)$ , 有  $(-\alpha, \beta) = 4 \neq -2 = -2(\alpha, \beta)$ .

(2) 不是, 取  $\alpha = (1, -1, 0, \dots, 0) \neq 0$ , 但  $(\alpha, \alpha) = 0$ .

(3) 是, 验证即可.

3. (1)  $(\beta, r) = X^T B Y$ , 其中  $X = (2, -3, 1)^T, Y = (-1, 1, -2)^T$

计算得  $(\beta, r) = -32$ .

(2)  $(\alpha_1, \alpha_2, \alpha_3) = (s_1, s_2, s_3) M$ , 其中  $M = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \Rightarrow M^{-1} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & -1 \\ -1 & 0 & 0 \end{pmatrix}$

故  $B = M^T C M$ ,  $C$  是  $s_1, s_2, s_3$  的度量矩阵.

$$C = (M^{-1})^T B M^{-1} = \begin{pmatrix} 3 & -5 & 3 \\ -5 & 8 & -4 \\ 3 & -4 & 3 \end{pmatrix}$$

4. Proof. 只需证  $|\alpha - r| \leq |\alpha - \beta| + |\beta - r|$

LHS =  $|\alpha - \beta + \beta - r| \leq$  RHS (三角不等式).

5. 展开即得.

6. (1) 易证. (2) 设  $g(x) = ax^2 + bx + c$ ,

$$\int_{-1}^1 g(x)(1+x) dx = \int_{-1}^1 ax^3 + (a+b)x^2 + (b+c)x + c dx$$

$$= \frac{a}{4}x^4 + \frac{a+b}{3}x^3 + \frac{b+c}{2}x^2 + cx \Big|_{-1}^1 = \frac{2}{3}(a+b) + 2c = 0$$

故  $g(x) = -3kx^2 + k$ .

类似地有  $\frac{2}{3}(a+b) + 2c = 0$  故  $\begin{cases} a = -3k \\ b = 0 \\ c = k \end{cases}$ .

取  $k = 1$ ,  $f(x) = 3x^2 - 1$  (不唯一).

7. (1)  $\alpha_1, \dots, \alpha_n$  是基, 则  $r = \sum_{i=1}^n k_i \alpha_i$ . 由  $(r, \alpha_i) = 0, i=1, 2, \dots, n$ .

$$\text{故 } (r, r) = \sum_{i=1}^n k_i (r, \alpha_i) = 0 \Leftrightarrow r = 0.$$

(2) 即  $\forall \alpha \in V$  有  $(r_1 - r_2, \alpha) = 0$ .

$$\text{取 } \alpha = r_1 - r_2 \Leftrightarrow r_1 - r_2 = 0 \text{ i.e. } r_1 = r_2$$

8. Schmidt 正交化即可. 长度定义:  $|\alpha| = \sqrt{(\alpha, \alpha)}$ .

9. 验证定义.

10. (1) 验证  $(\alpha_i, \alpha_j) = \begin{cases} 1, & i=j. \\ 0, & i \neq j. \end{cases}$  or 验证过渡矩阵是正交阵.

(2) 解 将  $\alpha_1, \alpha_2, \alpha_3$  正交化得

$$\beta_1 = \alpha_1 + \alpha_2,$$

$$\beta_2 = \alpha_2 - \frac{(\alpha_2, \beta_1)}{(\beta_1, \beta_1)} \beta_1 = \frac{1}{2}(\alpha_1 - 2\alpha_2 + 2\alpha_3 - \alpha_1),$$

$$\beta_3 = \alpha_3 - \frac{(\alpha_3, \beta_1)}{(\beta_1, \beta_1)} \beta_1 - \frac{(\alpha_3, \beta_2)}{(\beta_2, \beta_2)} \beta_2 = \alpha_1 + \alpha_2 + \alpha_3 - \alpha_3.$$

再单位化, 即得  $V_1$  的一组标准正交基:

$$\frac{1}{\sqrt{2}}(\alpha_1 + \alpha_2), \frac{1}{\sqrt{10}}(\alpha_1 - 2\alpha_2 + 2\alpha_3 - \alpha_1), \frac{1}{2}(\alpha_1 + \alpha_2 + \alpha_3 - \alpha_3).$$

11.  $\alpha = (\alpha_1 + c_1 i, \alpha_2 + c_2 i, \dots, \alpha_n + c_n i) \in \mathbb{C}^n$ .

$\beta = (b_1 + d_1 i, b_2 + d_2 i, \dots, b_n + d_n i) \in \mathbb{C}^n$ .

$$\text{定义 } (\alpha, \beta) = \sum_{i=1}^n \alpha_i b_i + \sum_{i=1}^n c_i d_i.$$

验证即可

## 补充题 8

1. 公众号内搜索往期推文.

2. 反证法. 假设  $\alpha_1, \alpha_2, \dots, \alpha_{n+2}$  两两正交.

$n$  维空间中任意  $n+1$  个向量线性相关, i.e.  $\exists$  不全为 0 的  $k_1, \dots, k_{n+1} \in \mathbb{R}$ ,

s.t.  $\sum_{i=1}^{n+1} k_i \alpha_i = 0$ . 两边同时内积  $\alpha_{n+2}$ , 有

$$\sum_{i=1}^{n+1} k_i (\alpha_i, \alpha_{n+2}) = 0, \text{ 且其中 } (\alpha_i, \alpha_{n+2}) \neq 0.$$

故存在  $k_{i_1}, k_{i_2}, \dots > 0; k_{j_1}, k_{j_2}, \dots < 0$  (零项忽略).

$$\text{故 } k_{i_1} \alpha_{i_1} + k_{i_2} \alpha_{i_2} + \dots = -k_{j_1} \alpha_{j_1} - k_{j_2} \alpha_{j_2} - \dots = v.$$

$$(v, v) = \sum -k_i k_j (\alpha_i, \alpha_j) < 0. \text{ 矛盾} \square$$

3. 必要性显然;

充分性:  $\alpha_i = 0 \cdot \alpha_1 + \dots + 1 \cdot \alpha_i + \dots + 0 \cdot \alpha_n$ .

$$\text{则 } (\alpha_i, \alpha_j) = \begin{cases} 1, & i=j, \\ 0, & i \neq j. \end{cases}$$

4. 必要性: 设  $\alpha_1, \dots, \alpha_n$  为标准正交基. 而

$$\alpha = \sum_{i=1}^n k_i \alpha_i \Rightarrow (\alpha, \alpha_i) = k_i$$

充分性: 设对  $\forall \alpha \in V$  均有  $\alpha = \sum_{i=1}^n (\alpha, \alpha_i) \alpha_i$

由于  $\alpha_i = 0 \alpha_1 + \dots + 1 \alpha_i + \dots + 0 \alpha_n$

$$= (\alpha_i, \alpha_i) \alpha_i + \dots + (\alpha_i, \alpha_i) \alpha_i + \dots + (\alpha_i, \alpha_n) \alpha_n$$

而在基下的表示是唯一的. 故  $(\alpha_i, \alpha_i) = 1, (\alpha_i, \alpha_j) = 0, i \neq j$ .

即  $\alpha_1, \dots, \alpha_n$  为标准正交基.

5.  $X = \sum_{i=1}^{n-s} t_i \beta_i$ , 有  $\alpha_j^T X = 0, j=1, 2, \dots, s$ .

取  $X = \beta_k, k=1, 2, \dots, n-s$ .

则  $\alpha_j$  与  $\beta_k$  线性无关, 否则  $\exists m \neq 0$ , s.t.  $\alpha_j = m\beta_k$

有  $m\beta_k^T \beta_k = 0 \Rightarrow \beta_k = 0$ , 矛盾.

6. Proof.  $\alpha \neq \beta$ . 故  $(\alpha - \beta, \alpha - \beta) > 0$

i.e.  $(\alpha, \alpha) - 2(\alpha, \beta) + (\beta, \beta) > 0$

即  $(\alpha, \beta) < 1$ . 故  $(\alpha, \beta) \neq 1$ .

7. 见补充题4, 这里不再赘述.